



Inspiring Excellence

CSE370: Database Systems Project Report

Project Title: Integrated Bus Transit Management System

Group No: 13, CSE370 Lab Section: 11, Spring 2026		
ID	Name	Contribution
23301107	Sheikh Srabon Ahmed	Worked on user management and ticket booking, including login, registration, seat selection, and ticket viewing.
23101211	Mushfique Nayeeb Ayon	Worked on route, bus stop, bus, and trip management, including the related database design and scheduling features.

22201877	Faysal Sheikh	Worked on payment, passenger tracking, and admin monitoring features, along with the supporting database operations.
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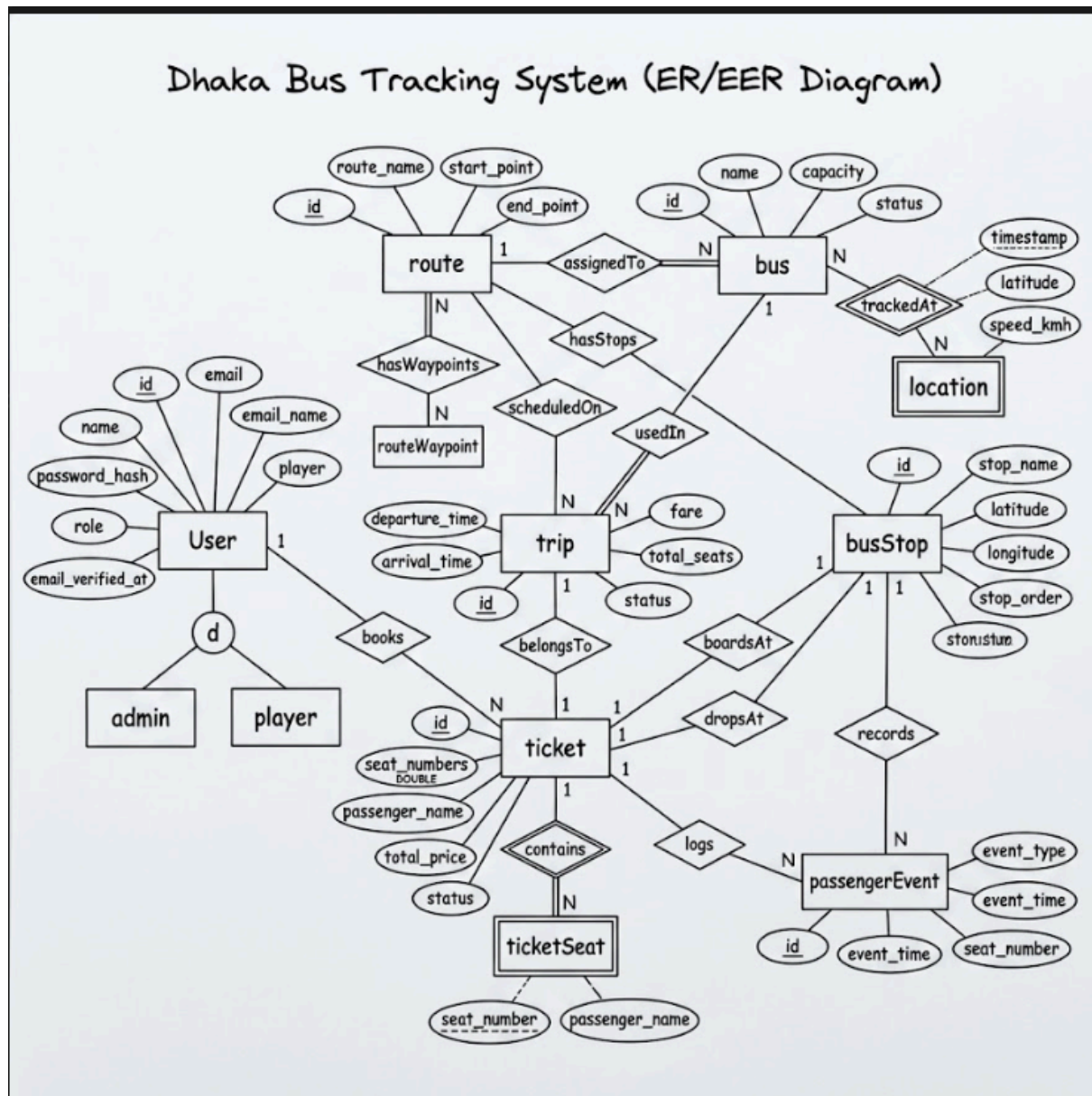
Introduction

Dhaka Bus is a database-driven web application designed to modernise urban bus ticketing, route management, and travel monitoring. The project integrates user registration, secure login, route and stop management, bus scheduling, seat-based ticket booking, payment processing, and administrative control within a single system. Its main objective is to demonstrate how a well-structured relational database can support real-time transport operations, maintain data consistency, and improve service efficiency. By combining backend logic, database design, and a responsive user interface, the system provides both passengers and administrators with an organised, practical, and scalable solution for digital bus transportation management in a smart city.

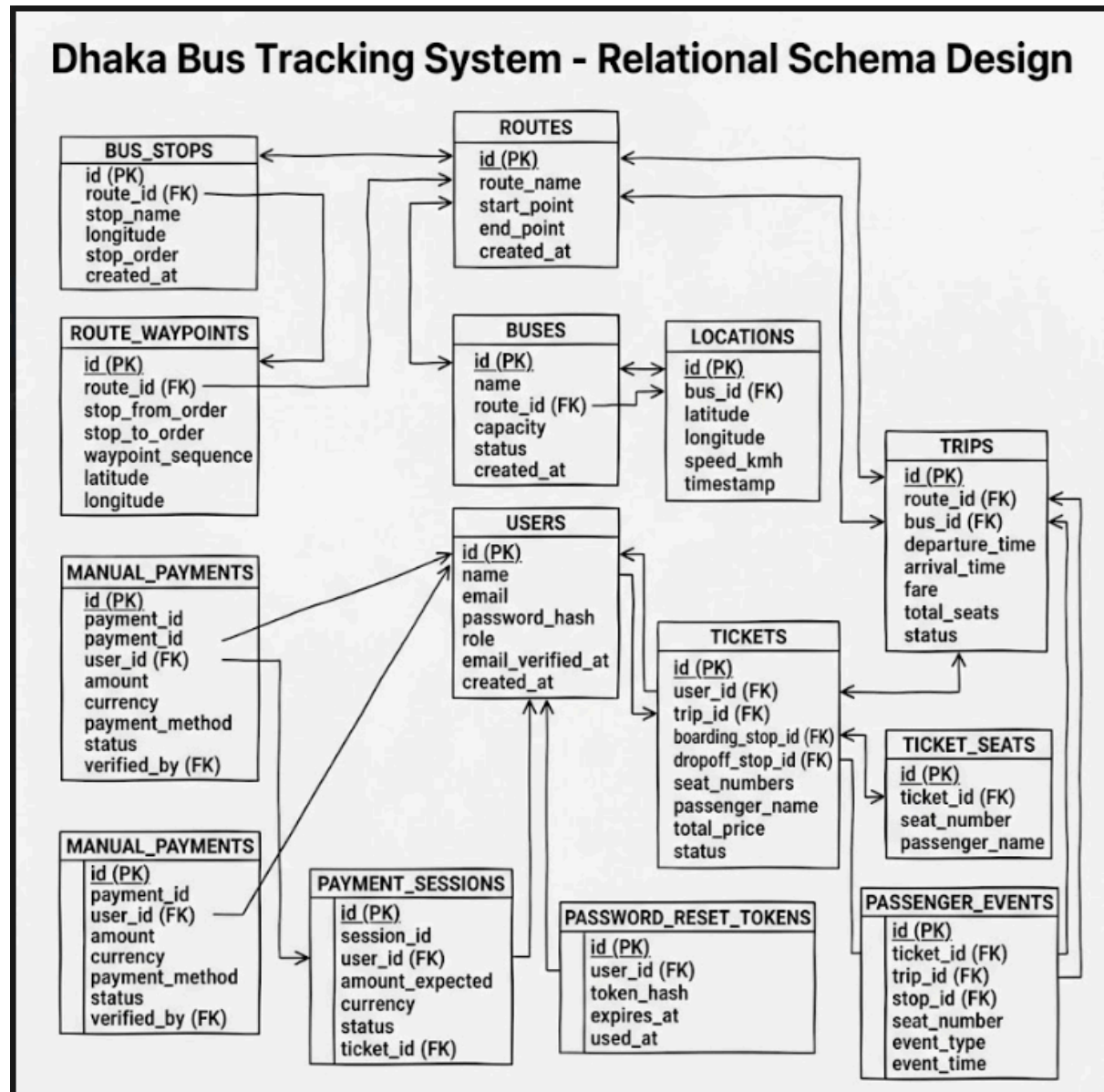
Project Features

ID, Name	Features [3 per member]	
23301107, Srabon	Ft 1	User registration and login system
	Ft 2	Seat-based ticket booking
	Ft 3	Ticket viewing and booking history
23101211, Ayon	Ft 1	Route and bus stop management
	Ft 2	Bus information management
	Ft 3	Trip scheduling and fare setup
22201877, Faysal	Ft 1	Payment processing and verification
	Ft 2	Passenger tracking and trip event monitoring
	Ft 3	Admin panel and report management

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

a. First Normal Form (1NF):

Yes, the converted schema is in 1NF. Each relation contains atomic attributes, and every field stores only a single value. There are no repeating groups or multivalued attributes in the final design. For example, seat-related information has been decomposed into separate relations such as ticket_seats and booking_request_seats, instead of storing multiple seat values in a single column. Therefore, the schema satisfies the requirements of 1NF.

b. Second Normal Form (2NF):

Yes, the converted schema is in 2NF. All non-key attributes are fully functionally dependent on the whole primary key. Since the relations mainly use single-attribute primary keys such as id, the possibility of partial functional dependency does not arise. Hence, there are no partial dependencies in the relational schema, and the design satisfies 2NF.

c. Third Normal Form (3NF):

Yes, the converted schema is in 3NF. The schema has been decomposed so that non-key attributes depend only on the primary key and not on other non-key attributes. Functional areas such as users, routes, stops, buses, trips, tickets, seat allocation, payments, and passenger events are stored in separate but related relations through foreign keys. This removes transitive dependency and ensures better consistency, reduced redundancy, and improved update efficiency. Therefore, the final converted schema satisfies 3NF.

Frontend Development

Briefly discuss about **Frontend** Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID: 23301107, **Name:** Srabon

Contributed to the design and implementation of the user-facing interfaces, including registration, login, booking workflow, seat selection, and ticket viewing pages. This work focused on usability, form interaction, and maintaining a clear step-by-step booking experience for users.

Contribution of ID: 23101211, **Name:** Ayon

Contributed to the interfaces related to transport management, including route, bus stop, bus, and trip scheduling pages. This included creating forms and display components for presenting operational transport data in an organized and accessible way.

Contribution of ID: 22201877, **Name:** Faysal

Contributed to the development of payment-related views, administrative pages, and monitoring interfaces. This included designing interface components that allowed administrative users to review tickets, payments, and system activities efficiently.

Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID: 23301107, Name : Srabon

Contributed to the development of authentication and booking-related server-side functionality, including user registration, login validation, role-based access support, and APIs for ticket booking, confirmation, and retrieval of ticket information.

Contribution of ID : 23101211, Name : Ayon

Contributed to the implementation of the core transport management logic, including APIs and backend processing for routes, bus stops, buses, trips, scheduling, and related availability functions connected to the database.

Contribution of ID : 22201877, Name : Faysal

Contributed to the implementation of payment processing, administrative APIs, ticket monitoring operations, and passenger tracking logic. This work supported booking verification, administrative control, and operational monitoring within the system.

Source Code Repository

<https://github.com/srabon007-9/dhaka-bus>

Conclusion

In conclusion, the Dhaka Bus project successfully demonstrates how a database-driven web application can be used to manage modern public transport operations in a structured and efficient manner. The system integrates important functions such as user authentication, route and trip management, seat-based ticket booking, payment handling, and administrative monitoring within a single platform. Through this project, the practical application of database concepts such as relational design, normalization, entity relationships, and query-based data management has been effectively implemented in a real-world context.

Furthermore, the project highlights the importance of combining frontend usability with backend processing and database integrity to build a complete software solution. It not only fulfills the academic objective of designing and implementing a database system, but also presents a scalable and practical model for digital transportation management. Overall, the project reflects both technical understanding and collaborative software development effort.

References

1. Express.js. (n.d.). Express - Node.js web application framework. Retrieved May 6, 2026, from <https://expressjs.com/>
2. MySQL. (n.d.). MySQL documentation. Retrieved May 6, 2026, from <https://dev.mysql.com/doc/>
3. Node.js. (n.d.). Node.js: Run JavaScript everywhere. Retrieved May 6, 2026, from <https://nodejs.org/en>
4. React. (n.d.). React. Retrieved May 6, 2026, from <https://react.dev/>
5. Socket.IO. (n.d.). Introduction. Retrieved May 6, 2026, from <https://socket.io/docs/v4/>
6. Vite. (n.d.). Getting started. Retrieved May 6, 2026, from <https://vite.dev/guide/>